# Workshop 2: DeepConvLSTM

ELEC 872

Labiba Farah and Hebatallah Sherif

**Setup.** We used OPPORTUNITY with 113 channels and 18 classes, keeping the supplied split and preprocessing. The reference has four convolutional layers, two LSTM layers, 24-sample windows and dropout 0.5. We trained on a Tesla T4 using RMSProp (learning rate 0.001), batch size 100, seed 0, and up to 100 epochs. Early-stopping patience was 20; checkpoints were selected by validation weighted F1.

| Run | Weighted F1 (%) | Macro F1 (%) | Params (M) | Time (min) |
|---|---|---|---|---|
| Reference | 88.73 | 59.12 | 3.97 | 17.78 |
| A: CNN | 86.29 | 44.61 | 7.49 | 13.11 |
| A: GRU | 89.38 | 60.93 | 2.99 | 12.40 |
| B: 1 convolution | 88.79 | 57.55 | 3.90 | 6.98 |
| **B: 2 convolutions** | **89.89** | **63.47** | 3.92 | 9.67 |
| C: window 20 | 89.08 | 58.14 | 3.97 | 9.88 |
| C: window 60 | 89.86 | 62.08 | 3.97 | 42.68 |
| D: baseline | 88.73 | 57.39 | 3.97 | 19.34 |
| D: no dropout | 88.81 | 58.61 | 3.97 | 20.37 |
| D: no weight decay | 88.96 | 57.62 | 3.97 | 19.42 |
| D: no augmentation | 88.43 | 56.57 | 3.97 | 17.47 |
| E: 5-epoch warm-up | 88.97 | 60.09 | 3.97 | 23.67 |
| F: focal loss | 89.44 | 58.73 | 3.97 | 30.65 |

**Part 0: Reference.** The reference reached 88.73% weighted F1 and 59.12% macro F1. Epoch 38 was selected; training stopped at epoch 58.

**Part A: CNN and GRU.** The CNN performed worse on both F1 scores. GRU improved both scores and used 24.6% fewer parameters than LSTM. The CNN has more parameters because its dense head receives the flattened features from all remaining time steps.

**Part B: Convolutions.** Two convolutional layers gave the best tested results: gains of 1.16 percentage points in weighted F1 and 4.35 points in macro F1 over the reference. More layers did not guarantee better scores.

**Part C: Window length.** The 60-sample window improved both scores but took 42.68 minutes, versus 17.78 minutes for the reference. The 20-sample window slightly improved weighted F1 while lowering macro F1.

**Part D: Regularization.** We used a separate D baseline with dropout 0.5, weight decay 0.0001 and augmentation enabled, since the original reference already had weight decay and augmentation off. Removing augmentation lowered both scores. Removing dropout or weight decay slightly improved test F1 in this run. However, the training/validation loss gap grew without dropout, consistent with stronger overfitting.

**Part E: Warm-up.** A five-epoch warm-up raised the learning rate from 0.0002 to 0.001. It slightly improved both F1 scores, but did not consistently smooth validation loss.

**Part F: Focal Loss.** Focal Loss ($\gamma = 2$, no class weights) improved weighted F1 but slightly lowered macro F1. Rare-gesture effects were mixed: Close Drawer 1 F1 rose from 0.406 to 0.440, while Close Drawer 2 fell from 0.400 to 0.310. Null makes up 83.27% of the reference test windows, so weighted F1 alone does not show whether rare gestures improved.

**Overall.** Two convolutions gave the highest observed F1 scores; GRU was an efficient alternative. These are single-seed results. Total training time also depends on how many epochs each run completed.

*Note:* Results and loss trends were recovered from the executed notebook's saved outputs; epoch losses were printed to four decimals. Sources: Ordóñez and Roggen (2016); Lin et al. (2017); OPPORTUNITY dataset.